



CHE 109: Engineering Chemistry – I
Summer-2025
Chemical solution of acids and bases
Chapter 2

Prof. Dr. Md. Kamal Hossain
Adjunct Faculty
East West University

► Colligative Properties

In the chapter opening, we mentioned that the addition of ethylene glycol, $\text{CH}_2\text{OHCH}_2\text{OH}$, to water lowers the freezing point of water below 0°C . For example, if 0.010 mol of ethylene glycol is added to 1 kg of water, the freezing point is lowered to -0.019°C . The magnitude of freezing-point lowering is directly proportional to the number of ethylene glycol molecules added to a quantity of water. Thus, if you add 0.020 ($=0.010 \times 2$) mol of ethylene glycol to 1 kg of water, the freezing point is lowered to -0.038°C ($=-0.019^\circ\text{C} \times 2$). The same lowering is observed with the addition of other nonelectrolyte substances. For example, an aqueous solution of 0.020 mol of urea, $(\text{NH}_2)_2\text{CO}$, in 1 kg of water also freezes at -0.038°C .

Freezing-point lowering is a colligative property. **Colligative properties** of solutions are *properties that depend on the concentration of solute molecules or ions in solution but not on the chemical identity of the solute* (whether it is ethylene glycol or urea, for instance). In the next sections, we will discuss several colligative properties, which are expressed quantitatively in terms of various concentration units. We will first look into ways of expressing the concentration of a solution.

12.4 Ways of Expressing Concentration

The *concentration* of a solute is the amount of solute dissolved in a given quantity of solvent or solution. The quantity of solvent or solution can be expressed in terms of volume or in terms of mass or molar amount. Thus, there are several ways of expressing the concentration of a solution.

Recall that the *molarity* of a solution is the moles of solute in a liter of solution. ►

$$\text{Molarity} = \frac{\text{moles of solute}}{\text{liters of solution}}$$

For example, 0.20 mol of ethylene glycol dissolved in enough water to give 2.0 L of solution has a molarity of

$$\frac{0.20 \text{ mol ethylene glycol}}{2.0 \text{ L solution}} = 0.10 \text{ } M \text{ ethylene glycol}$$

The solution is 0.10 molar (denoted M). This unit is especially useful when you wish to dispense a given amount of solute, because the unit directly relates the amount of solute to the volume of solution.

Some other concentration units are defined in terms of the mass or molar amount of solvent or solution. The most important of these are mass percentage of solute, molality, and mole fraction.

Mass Percentage of Solute

Solution concentration is sometimes expressed in terms of the **mass percentage of solute**—that is, *the percentage by mass of solute contained in a solution*.

$$\text{Mass percentage of solute} = \frac{\text{mass of solute}}{\text{mass of solution}} \times 100\%$$

For example, an aqueous solution that is 3.5% sodium chloride by mass contains 3.5 g of NaCl in 100.0 g of solution. It could be prepared by dissolving 3.5 g of NaCl in 96.5 g of water ($100.0 - 3.5 = 96.5$).

Example 12.2 Calculating with Mass Percentage of Solute

Gaining Mastery Toolbox

Critical Concept 12.2

Solution concentration can be expressed as a mass percentage of solute. When expressing solution concentration as a mass percent, the units of mass for the solute and solvent must be the same, for example, grams of solute and grams of solution. The mass of solution is the sum of the mass(es) of the solute(s) and the solvent.

Solution Essentials:

- Mass percentage of solute
(% mass = $\frac{\text{mass of solute}}{\text{mass of solution}} \times 100\%$)
- Solute
- Solution
- Concentration

How would you prepare 425 g of an aqueous solution containing 2.40% by mass of sodium acetate, $\text{NaC}_2\text{H}_3\text{O}_2$?

Problem Strategy In order to know how to prepare this solution, we need to know the mass of solute ($\text{NaC}_2\text{H}_3\text{O}_2$) and the mass of solution. The mass percent given in the problem (2.40%) will enable us to determine the mass of solute present in the 425-g sample. Keeping in mind that the total mass of solution equals the sum of the mass of solute and the mass of solvent (mass solution = mass solute + mass solvent), we will be able to determine the mass of H_2O needed to prepare the solution.

Solution The mass of sodium acetate (solute) in 425 g of solution is

$$\text{Mass of NaC}_2\text{H}_3\text{O}_2 = 425 \text{ g} \times 0.0240 = 10.2 \text{ g}$$

We have the relationship

$$\text{Mass of solution} = \text{mass NaC}_2\text{H}_3\text{O}_2 + \text{mass H}_2\text{O}$$

Therefore, the quantity of water in the solution is

$$\text{Mass of H}_2\text{O} = \text{mass of solution} - \text{mass of NaC}_2\text{H}_3\text{O}_2 = 425 \text{ g} - 10.2 \text{ g} = 415 \text{ g}$$

You would prepare the solution by dissolving 10.2 g of sodium acetate in 415 g of water.

Molality

The **molality** of a solution is *the moles of solute per kilogram of solvent*.

$$\text{Molality} = \frac{\text{moles of solute}}{\text{kilograms of solvent}}$$

For example, 0.20 mol of ethylene glycol dissolved in $2.0 \times 10^3 \text{ g}$ ($=2.0 \text{ kg}$) of water has a molality of

$$\frac{0.20 \text{ mol ethylene glycol}}{2.0 \text{ kg solvent}} = 0.10 \text{ } m \text{ ethylene glycol}$$

That is, the solution is 0.10 molal (denoted *m*). The units of molality and molarity are sometimes confused. Note that molality is defined in terms of *mass of solvent*, and molarity is defined in terms of *volume of solution*.

Example 12.3 Calculating the Molality of Solute

Gaining Mastery Toolbox

Critical Concept 12.3

Solution concentration can be expressed as the ratio of the moles of solute to the mass of solvent: molality (*m*). When calculating solution molality, the mass of solute must be expressed in kg.

Solution Essentials:

• Molality

$$\left(m = \frac{\text{moles of solute}}{\text{kg of solvent}} \right)$$

• Solute

• Solvent

• Concentration

Glucose, $\text{C}_6\text{H}_{12}\text{O}_6$, is a sugar that occurs in fruits. It is also known as “blood sugar” because it is found in blood and is the body’s main source of energy (see Figure 12.16). What is the molality of a solution containing 5.67 g of glucose dissolved in 25.2 g of water?

Problem Strategy Recall that the definition of *molality* is moles of solute per kilogram of solvent. Therefore, we need to determine these two quantities to solve this problem. Start by converting mass of solute (glucose) to moles, and next determine the mass of solvent (water) in kilograms. Then use these quantities to calculate the solution molality.

Solution The moles of glucose ($\text{MM} = 180.2 \text{ amu}$) in 5.67 g are found as follows:

$$5.67 \text{ g } \cancel{\text{C}_6\text{H}_{12}\text{O}_6} \times \frac{1 \text{ mol } \text{C}_6\text{H}_{12}\text{O}_6}{180.2 \text{ g } \cancel{\text{C}_6\text{H}_{12}\text{O}_6}} = 0.0315 \text{ mol } \text{C}_6\text{H}_{12}\text{O}_6$$

The mass of water is 25.2 g, or $25.2 \times 10^{-3} \text{ kg}$.

$$\text{Molality} = \frac{0.0315 \text{ mol } \text{C}_6\text{H}_{12}\text{O}_6}{25.2 \times 10^{-3} \text{ kg solvent}} = 1.25 \text{ } m \text{ } \text{C}_6\text{H}_{12}\text{O}_6$$

Mole Fraction

The **mole fraction** of a component substance A (X_A) in a solution is defined as *the moles of component substance divided by the total moles of solution* (that is, moles of solute plus solvent).

$$X_A = \frac{\text{moles of substance } A}{\text{total moles of solution}}$$

For example, if a solution is made up of 1 mol of ethylene glycol and 9 mol of water, the total moles of solution are 1 mol + 9 mol = 10 mol. The mole fraction of ethylene glycol is $1/10 = 0.1$, and the mole fraction of water is $9/10 = 0.9$. Multiplying mole fractions by 100 gives *mole percent*. Hence, this solution is 10 mole percent ethylene glycol and 90 mole percent water. You can also say that 10% of the molecules in the solution are ethylene glycol and 90% are water. The sum of the mole fractions of all the components of a solution equals 1.

Conversion of Concentration Units

It is relatively easy to interconvert concentration units when they are expressed in terms of mass or moles of solute and solvent, as the following examples show.

Example 12.5 Converting Molality to Mole Fractions

Gaining Mastery Toolbox

Critical Concept 12.5

The concentration of a solution can be expressed in many different ways. Given the concentration of a solution in one set of units, it is often possible to convert to a different set of units. It is critical to keep track of all of the components that define a solution (solute, solvent, and solution) because the concentration units are defined using these quantities. In order to solve the problem in Example 12.5, we must first determine the moles of solute (glucose), moles of solvent (water), and moles of solution (glucose and water).

Solution Essentials:

- Molality $\left(m = \frac{\text{moles of solute}}{\text{kg of solvent}}\right)$
- Mole fraction $\left(x_A = \frac{\text{moles of substance A}}{\text{total moles of solution}}\right)$
- Solute
- Solvent
- Solution

An aqueous solution is 0.120 *m* glucose, $\text{C}_6\text{H}_{12}\text{O}_6$. What are the mole fractions of each component in the solution?

Problem Strategy Because molality is the moles of solute per kilogram of solvent, a 0.120 *m* glucose solution contains 0.120 mol of glucose in 1.00 kg of water. After converting 1.00 kg H_2O to moles, you can calculate the mole fraction using its definition.

Solution The moles of H_2O in 1.00 kg of water are

$$1.00 \times 10^3 \text{ g } \cancel{\text{H}_2\text{O}} \times \frac{1 \text{ mol } \text{H}_2\text{O}}{18.0 \text{ g } \cancel{\text{H}_2\text{O}}} = 55.6 \text{ mol } \text{H}_2\text{O}$$

Hence,

$$\text{Mole fraction glucose} = \frac{0.120 \text{ mol}}{(0.120 + 55.6) \text{ mol}} = 0.00215$$

$$\text{Mole fraction water} = \frac{55.6 \text{ mol}}{(0.120 + 55.6) \text{ mol}} = 0.998$$

Answer Check Check to see that the mole fractions of each substance sum to 1.

Exercise 12.8 A solution is 0.120 *m* methanol dissolved in ethanol. Calculate the mole fractions of methanol, CH_3OH , and ethanol, $\text{C}_2\text{H}_5\text{OH}$, in the solution.

See Problems 12.59 and 12.60.

Example 12.6 Converting Mole Fractions to Molality

Gaining Mastery Toolbox

Critical Concept 12.6

The concentration of a solution can be expressed in many different ways. Given the concentration of a solution in one set of units, it is often possible to convert to a different set of units. It is critical to keep track of all of the components that define a solution (solute, solvent, and solution) because the concentration units are defined using these quantities. The solution to the problem in Example 12.6 requires that we first determine the moles of solute (glucose) and the mass of solvent (water).

Solution Essentials:

• Molality

$$m = \frac{\text{moles of solute}}{\text{kg of solvent}}$$

• Mole fraction

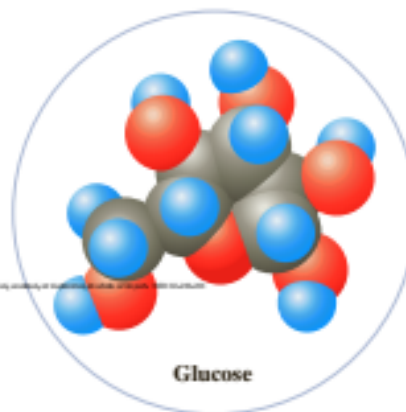
$$X_A = \frac{\text{moles of substance A}}{\text{total moles of solution}}$$

• Solute

• Solvent

• Solution

A solution is 0.150 mole fraction glucose, $\text{C}_6\text{H}_{12}\text{O}_6$, and 0.850 mole fraction water. What is the molality of glucose in the solution?



Problem Strategy The molality of the solution is the moles of solute (glucose) per kilogram of solvent (H_2O). Given the mole fraction information in the problem, we know that each mole of solution contains 0.150 mol of glucose and 0.850 mol of water. If we convert the moles of water (0.850 mol H_2O) to mass of water, in kilograms, we can then calculate the molality.

Solution The mass of this amount of water is

$$0.850 \text{ mol } \text{H}_2\text{O} \times \frac{18.0 \text{ g } \text{H}_2\text{O}}{1 \text{ mol } \text{H}_2\text{O}} = 15.3 \text{ g } \text{H}_2\text{O} (= 0.0153 \text{ kg } \text{H}_2\text{O}) \quad (\text{continued})$$

Example 12.6 (continued)

The molality of glucose, $\text{C}_6\text{H}_{12}\text{O}_6$, in the solution is

$$\text{Molality of } \text{C}_6\text{H}_{12}\text{O}_6 = \frac{0.150 \text{ mol } \text{C}_6\text{H}_{12}\text{O}_6}{0.0153 \text{ kg solvent}} = 9.80 \text{ m } \text{C}_6\text{H}_{12}\text{O}_6$$

Example 12.7 Converting Molality to Molarity

Gaining Mastery Toolbox

Critical Concept 12.7

The concentration of a solution can be expressed in many different ways. Given the concentration of a solution in one set of units, it is often possible to convert to a different set of units. It is critical to keep track of all of the components that define a solution (solute, solvent, and solution) because the concentration units are defined using these quantities. The solution to the problem in Example 12.7 requires that we first determine the mass of solute (KCl) and the mass of solution (KCl and water).

Solution Essentials:

• Molality

$$m = \frac{\text{moles of solute}}{\text{kg of solvent}}$$

• Molarity

$$M = \frac{\text{moles of solute}}{\text{L of solution}}$$

• Solute

• Solvent

• Solution

An aqueous solution is 0.273 *m* KCl. What is the molar concentration of potassium chloride, KCl? The density of the solution is $1.011 \times 10^3 \text{ g/L}$.

Problem Strategy You are asked to determine the molarity of the solution, which is the moles of solute (KCl) per liter of solution (volume of H₂O and KCl). The molality tells you that the solution contains 0.273 mol KCl in 1 kg of water. If we determine the volume of solution that contains 0.273 mol KCl, we should be able to calculate the molarity. Because the solution is made up of only KCl and water, the solution mass is the mass of 0.273 mol KCl plus the mass of 1.000 kg of water. Knowing the total mass of solution, we then can use the solution density to convert from the mass of solution to the volume of solution. The molarity is obtained by dividing the moles of solute by the volume of solution in liters.

Solution The first step in determining the volume of solution is to find the mass of potassium chloride in this quantity of solution:

$$0.273 \text{ mol KCl} \times \frac{74.6 \text{ g KCl}}{1 \text{ mol KCl}} = 20.4 \text{ g KCl}$$

The total mass of the solution equals the mass of water plus the mass of potassium chloride.

$$1.000 \times 10^3 \text{ g} + 20.4 \text{ g} = 1.020 \times 10^3 \text{ g}$$

The volume of the solution can be calculated using the inverse of the density of the solution.

$$\text{Volume of solution} = 1.020 \times 10^3 \text{ g} \times \frac{1 \text{ L}}{1.011 \times 10^3 \text{ g}} = 1.009 \text{ L}$$

There is 0.273 mol KCl in every 1.009 L of solution. Hence, the molarity of the solution is

$$\frac{0.273 \text{ mol KCl}}{1.009 \text{ L solution}} = 0.271 \text{ M KCl}$$

Note that the molarity and the molality of this solution are approximately equal. This happens in cases where the solutions are dilute and the density is about 1 g/mL.

Example 12.8 Converting Molarity to Molality

Gaining Mastery Toolbox

Critical Concept 12.8

The concentration of a solution can be expressed in many different ways. Given the concentration of a solution in one set of units, it is often possible to convert to a different set of units. It is critical to keep track of all of the components that define a solution (solute, solvent, and solution) because the concentration units are defined using these quantities. When solving the problem in Example 12.8, it helps to assume that you have a 1-L sample of the solution.

Solution Essentials:

- Molality

$$m = \frac{\text{moles of solute}}{\text{kg of solvent}}$$

- Molarity

$$M = \frac{\text{moles of solute}}{\text{L of solution}}$$

- Solute
- Solvent
- Solution

An aqueous solution is 0.907 *M* $\text{Pb}(\text{NO}_3)_2$. What is the molality of lead(II) nitrate, $\text{Pb}(\text{NO}_3)_2$, in this solution? The density of the solution is 1.252 g/mL.

Problem Strategy You are asked to calculate the molality of the solution, which is the moles of solute ($\text{Pb}(\text{NO}_3)_2$) per kilogram of solvent (H_2O). From the molarity, we know that there are 0.907 mol $\text{Pb}(\text{NO}_3)_2$ per liter of solution. If we consider a 1-L sample of this solution, we can use the density to determine the mass of solution. Because the mass of solution is the mass of lead(II) nitrate plus the mass of water, you can calculate the mass of lead(II) nitrate by difference (mass H_2O = mass of solution – mass of $\text{Pb}(\text{NO}_3)_2$). Now you can calculate the molality.

Solution Assuming 1.000 L (1.000×10^3 mL) of solution,

$$\begin{aligned}\text{Mass of solution} &= \text{density} \times \text{volume} \\ &= 1.252 \text{ g/mL} \times 1.000 \times 10^3 \text{ mL} \\ &= 1.252 \times 10^3 \text{ g}\end{aligned}$$

The mass of lead nitrate is

$$0.907 \text{ mol } \text{Pb}(\text{NO}_3)_2 \times \frac{331.2 \text{ g } \text{Pb}(\text{NO}_3)_2}{1 \text{ mol } \text{Pb}(\text{NO}_3)_2} = 3.00 \times 10^2 \text{ g } \text{Pb}(\text{NO}_3)_2$$

The mass of water in 1.000 L of solution is

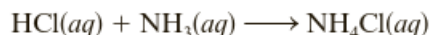
$$\begin{aligned}\text{Mass of } \text{H}_2\text{O} &= \text{mass of solution} - \text{mass of } \text{Pb}(\text{NO}_3)_2 \\ &= 1.252 \times 10^3 \text{ g} - 3.00 \times 10^2 \text{ g} \\ &= 9.52 \times 10^2 \text{ g} (= 0.952 \text{ kg } \text{H}_2\text{O})\end{aligned}$$

Hence, the molality of lead(II) nitrate in this solution is

$$\frac{0.907 \text{ mol } \text{Pb}(\text{NO}_3)_2}{0.952 \text{ kg solvent}} = 0.953 \text{ } m \text{ } \text{Pb}(\text{NO}_3)_2$$

Acids and bases were first recognized by simple properties, such as taste. Acids have a sour taste; bases are bitter. Also, acids and bases change the color of certain dyes called indicators, such as litmus and phenolphthalein. Acids change litmus from blue to red and basic phenolphthalein from red to colorless. Bases change litmus from red to blue and phenolphthalein from colorless to red. As you can see from these color changes, acids and bases neutralize, or reverse, the action of one another. During neutralization, acids and bases react with each other to produce ionic substances called salts. Acids react with active metals, such as magnesium and zinc, to release hydrogen.

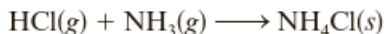
The Swedish chemist Svante Arrhenius framed the first successful concept of acids and bases. He defined acids and bases in terms of the effect these substances have on water. According to Arrhenius, acids are substances that increase the concentration of H^+ ion in aqueous solution, and bases increase the concentration of OH^- ion in aqueous solution. But many reactions that have characteristics of acid–base reactions in aqueous solution occur in other solvents or without a solvent. For example, hydrochloric acid reacts with aqueous ammonia, which in the Arrhenius view is a base because it increases the concentration of OH^- ion in aqueous solution. The reaction can be written



The product is a solution of NH_4Cl —that is, a solution of NH_4^+ and Cl^- ions. A very similar reaction occurs between hydrogen chloride and ammonia dissolved in benzene, C_6H_6 . The product is again NH_4Cl , which in this case precipitates from the solution.



Hydrogen chloride and ammonia react even in the gas phase. If watch glasses (shallow glass dishes) of concentrated hydrochloric acid and concentrated ammonia are placed next to each other, dense, white fumes of NH_4Cl particles form where HCl gas and NH_3 gas come into contact (Figure 15.1).



These reactions of HCl and NH_3 , in benzene and in the gas phase, are similar to the reaction in aqueous solution but cannot be explained by the Arrhenius concept. Broader acid–base concepts are needed.



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Figure 15.1 ▲

Reaction of $\text{HCl}(g)$ and $\text{NH}_3(g)$ to form $\text{NH}_4\text{Cl}(s)$ Gases from the concentrated solutions diffuse from their watch glasses (shallow dishes) and react to give a smoke of ammonium chloride particles.

15.1 Arrhenius Concept of Acids and Bases

We can state the Arrhenius concept of an acid as follows: *An acid is a substance that, when dissolved in water, increases the concentration of hydronium ion, $\text{H}_3\text{O}^+(\text{aq})$.* For simplicity, chemists frequently use the notation $\text{H}^+(\text{aq})$ for the $\text{H}_3\text{O}^+(\text{aq})$ ion and call

This section summarizes the discussion in section 4.4.

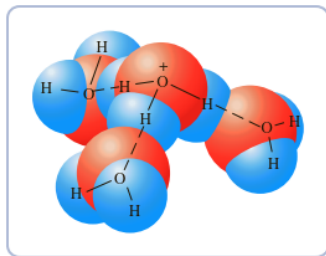


Figure 15.2 ▲

The hydronium ion, H_3O^+ The species is shown here hydrogen-bonded to three water molecules. The positive charge shown is actually distributed over the ion.

We introduced strong and weak acids and bases in Section 4.4.

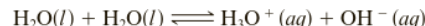
Table 15.1 Common Strong Acids and Bases

Strong Acids	Strong Bases*
HClO_4	LiOH
H_2SO_4	NaOH
HI	KOH
HBr	$\text{Ca}(\text{OH})_2$
HCl	$\text{Sr}(\text{OH})_2$
HNO_3	$\text{Ba}(\text{OH})_2$

*In general, the Group 1A and Group 2A hydroxides (except beryllium hydroxide) are strong bases.

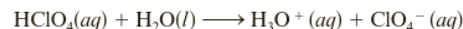
it the hydrogen ion. Remember, however, that the aqueous hydrogen ion is not a bare proton in water, but a proton chemically bonded to water—that is, $\text{H}_3\text{O}^+(\text{aq})$. The $\text{H}_3\text{O}^+(\text{aq})$ ion is itself associated through hydrogen bonding with a variable number of water molecules. One such species is shown in Figure 15.2. *A base, in the Arrhenius concept, is a substance that, when dissolved in water, increases the concentration of hydroxide ion, $\text{OH}^-(\text{aq})$.* ◀

The special role of the hydronium ion (or hydrogen ion) and the hydroxide ion in aqueous solutions arises from the following reaction:



The addition of acids and bases alters the concentrations of these ions in water.

In Arrhenius's theory, a *strong acid* is a substance that completely ionizes in aqueous solution to give $\text{H}_3\text{O}^+(\text{aq})$ and an anion. An example is perchloric acid, HClO_4 .



Other examples of strong acids are H_2SO_4 , HI , HBr , HCl , and HNO_3 . A *strong base* completely ionizes in aqueous solution to give OH^- and a cation. Sodium hydroxide is an example of a strong base.

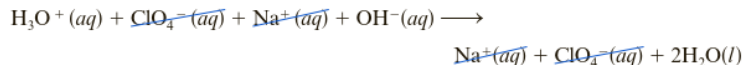


The principal strong bases are the hydroxides of Group 1A elements and Group 2A elements (except Be). See Table 15.1. ◀

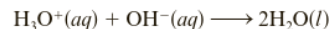
Most of the other acids and bases you encounter are *weak*. They are not completely ionized in solution and exist in reversible reaction with the corresponding ions. Consider acetic acid, $\text{HC}_2\text{H}_3\text{O}_2$. The reaction with water is



Evidence for the Arrhenius theory comes from the heat of reaction, ΔH° , for the neutralization of a strong acid by a strong base. This neutralization is essentially the reaction of $\text{H}_3\text{O}^+(\text{aq})$ with $\text{OH}^-(\text{aq})$ and should therefore always give the same ΔH° per mole of water formed. For example, if you write the neutralization of HClO_4 with NaOH in ionic form, you have



After canceling, you get the net ionic equation



Experimentally, it is found that all neutralizations involving strong acids and bases have the same ΔH° : $-55.90 \text{ kJ per mole of } \text{H}^+$. This indicates that the same reaction occurs in each neutralization, as Arrhenius's theory predicts.

Despite its successes, the Arrhenius concept is limited. In addition to looking at acid–base reactions only in aqueous solutions, it singles out the OH^- ion as the source of base character, when other species can play a similar role. Broader definitions of acids and bases are described in the next sections.

15.1 Brønsted Acids and Bases

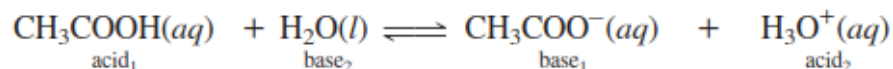
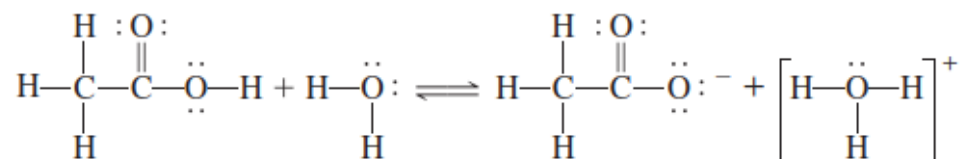
In Chapter 4 we defined a Brønsted acid as a substance capable of donating a proton, and a Brønsted base as a substance that can accept a proton. These definitions are generally suitable for a discussion of the properties and reactions of acids and bases.

An extension of the Brønsted definition of acids and bases is the concept of the **conjugate acid-base pair**, which can be defined as *an acid and its conjugate base or a base and its conjugate acid*. The conjugate base of a Brønsted acid is the species that remains when one proton has been removed from the acid. Conversely, a conjugate acid results from the addition of a proton to a Brønsted base.

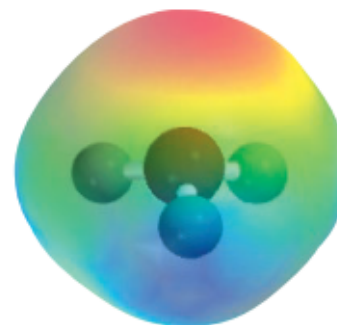
Every Brønsted acid has a conjugate base, and every Brønsted base has a conjugate acid. For example, the chloride ion (Cl^-) is the conjugate base formed from the acid HCl , and H_3O^+ (hydronium ion) is the conjugate acid of the base H_2O .



Similarly, the ionization of acetic acid can be represented as



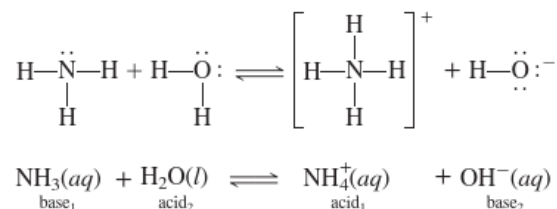
Conjugate means "joined together."



The proton is always associated with water molecules in aqueous solution. The H_3O^+ ion is the simplest formula of a hydrated proton.

The subscripts 1 and 2 designate the two conjugate acid-base pairs. Thus, the acetate ion (CH_3COO^-) is the conjugate base of CH_3COOH . Both the ionization of HCl (see Section 4.3) and the ionization of CH_3COOH are examples of Brønsted acid-base reactions.

The Brønsted definition also enables us to classify ammonia as a base because of its ability to accept a proton:



Student Hot Spot

Student data indicate you may struggle with conjugate acid-base pairs. Access your eBook for additional Learning Resources on this topic.

In this case, NH_4^+ is the conjugate acid of the base NH_3 , and the hydroxide ion OH^- is the conjugate base of the acid H_2O . Note that the atom in the Brønsted base that accepts a H^+ ion must have a lone pair.

In Example 15.1, we identify the conjugate pairs in an acid-base reaction.

Example 15.1

Identify the conjugate acid-base pairs in the reaction between ammonia and hydrofluoric acid in aqueous solution



Strategy Remember that a conjugate base always has one fewer H atom and one more negative charge (or one fewer positive charge) than the formula of the corresponding acid.

Solution NH_3 has one fewer H atom and one fewer positive charge than NH_4^+ . F^- has one fewer H atom and one more negative charge than HF . Therefore, the conjugate acid-base pairs are (1) NH_4^+ and NH_3 and (2) HF and F^- .

Practice Exercise Identify the conjugate acid-base pairs for the reaction

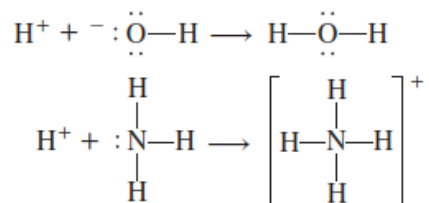


Similar problem: 15.5.

It is acceptable to represent the proton in aqueous solution either as H^+ or as H_3O^+ . The formula H^+ is less cumbersome in calculations involving hydrogen ion concentrations and in calculations involving equilibrium constants, whereas H_3O^+ is more useful in a discussion of Brønsted acid-base properties.

15.12 Lewis Acids and Bases

So far we have discussed acid-base properties in terms of the Brønsted theory. To behave as a Brønsted base, for example, a substance must be able to accept protons. By this definition both the hydroxide ion and ammonia are bases:



In each case, the atom to which the proton becomes attached possesses at least one unshared pair of electrons. This characteristic property of OH^- , NH_3 , and other Brønsted bases suggests a more general definition of acids and bases.

In 1932 the American chemist G. N. Lewis formulated such a definition. He defined what we now call a **Lewis base** as a substance that can donate a pair of electrons. A **Lewis acid** is a substance that can accept a pair of electrons. For example, in the protonation of ammonia, NH_3 acts as a Lewis base because it donates a pair of electrons to the proton H^+ , which acts as a Lewis acid by accepting the pair of electrons. A Lewis acid-base reaction, therefore, is one that involves the donation of a pair of electrons from one species to another. Such a reaction does not produce a salt and water.

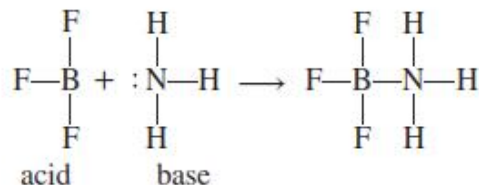
Lewis acids are either deficient in electrons (cations) or the central atom has a vacant valence orbital.



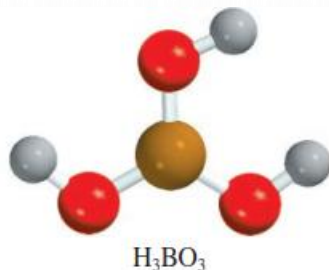
Figure 15.11 A Lewis acid-base reaction involving BF_3 and NH_3 .
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A coordinate covalent bond (see Section 9.9) is always formed in a Lewis acid-base reaction.

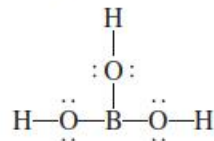
The significance of the Lewis concept is that it is more general than other definitions. Lewis acid-base reactions include many reactions that do not involve Brønsted acids. Consider, for example, the reaction between boron trifluoride (BF_3) and ammonia to form an adduct compound (Figure 15.11):



In Section 10.4 we saw that the B atom in BF_3 is sp^2 -hybridized. The vacant, unhybridized $2p_z$ orbital accepts the pair of electrons from NH_3 . So BF_3 functions as an acid according to the Lewis definition, even though it does not contain an ionizable proton. Note that a coordinate covalent bond is formed between the B and N atoms, as is the case in all Lewis acid-base reactions.



Another Lewis acid containing boron is boric acid. Boric acid (a weak acid used in eyewash) is an oxoacid with the following structure:



Boric acid does not ionize in water to produce a H^+ ion. Its reaction with water is

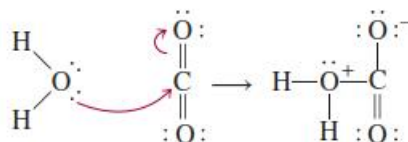


In this Lewis acid-base reaction, boric acid accepts a pair of electrons from the hydroxide ion that is derived from the H_2O molecule.

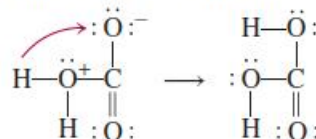
The hydration of carbon dioxide to produce carbonic acid



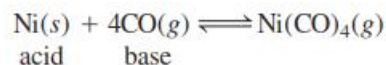
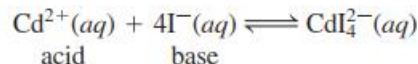
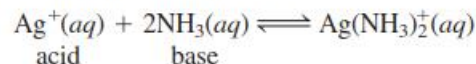
can be understood in the Lewis framework as follows: The first step involves donation of a lone pair on the oxygen atom in H_2O to the carbon atom in CO_2 . An orbital is vacated on the C atom to accommodate the lone pair by removal of the electron pair in the C—O pi bond. These shifts of electrons are indicated by the curved arrows.



Therefore, H_2O is a Lewis base and CO_2 is a Lewis acid. Next, a proton is transferred onto the O atom bearing a negative charge to form H_2CO_3 .

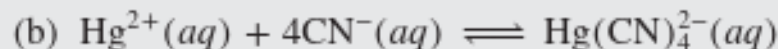
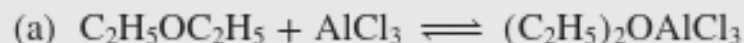


Other examples of Lewis acid-base reactions are



Example 15.15

Identify the Lewis acid and Lewis base in each of the following reactions:

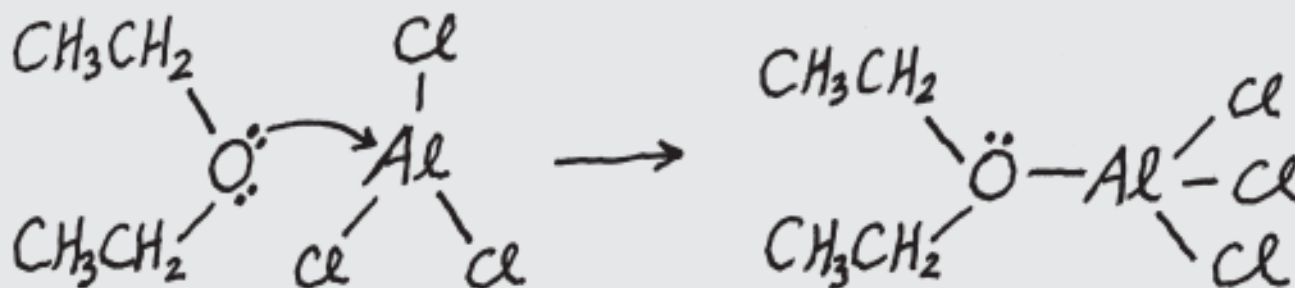


Strategy In Lewis acid-base reactions, the acid is usually a cation or an electron-deficient molecule, whereas the base is an anion or a molecule containing an atom with lone pairs.

(a) Draw the molecular structure for $\text{C}_2\text{H}_5\text{OC}_2\text{H}_5$. What is the hybridization state of Al in AlCl_3 ? (b) Which ion is likely to be an electron acceptor? An electron donor?

Solution

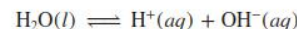
- (a) The Al is sp^2 -hybridized in AlCl_3 with an empty $2p_z$ orbital. It is electron-deficient, sharing only six electrons. Therefore, the Al atom has a tendency to gain two electrons to complete its octet. This property makes AlCl_3 a Lewis acid. On the other hand, the lone pairs on the oxygen atom in $\text{C}_2\text{H}_5\text{OC}_2\text{H}_5$ make the compound a Lewis base:



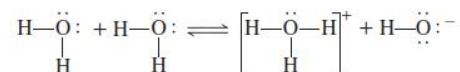
- (b) Here the Hg^{2+} ion accepts four pairs of electrons from the CN^{-} ions. Therefore, Hg^{2+} is the Lewis acid and CN^{-} is the Lewis base.

15.2 The Acid-Base Properties of Water

Water, as we know, is a unique solvent. One of its special properties is its ability to act either as an acid or as a base. Water functions as a base in reactions with acids such as HCl and CH₃COOH, and it functions as an acid in reactions with bases such as NH₃. Water is a very weak electrolyte and therefore a poor conductor of electricity, but it does undergo ionization to a small extent:



This reaction is sometimes called the *autoionization* of water. To describe the acid-base properties of water in the Brønsted framework, we express its autoionization as follows (also shown in Figure 15.1):



The acid-base conjugate pairs are (1) H₂O (acid) and OH[−] (base) and (2) H₃O⁺ (acid) and H₂O (base).

The Ion Product of Water

In the study of acid-base reactions, the hydrogen ion concentration is key; its value indicates the acidity or basicity of the solution. Because only a very small fraction of water molecules are ionized, the concentration of water, [H₂O], remains virtually unchanged. Therefore, the equilibrium constant for the autoionization of water, according to Equation (15.1), is

$$K_c = [\text{H}_3\text{O}^+][\text{OH}^-]$$

Because we use H⁺(aq) and H₃O⁺(aq) interchangeably to represent the hydrated proton, the equilibrium constant can also be expressed as

$$K_c = [\text{H}^+][\text{OH}^-]$$

To indicate that the equilibrium constant refers to the autoionization of water, we replace K_c with K_w

$$K_w = [\text{H}_3\text{O}^+][\text{OH}^-] = [\text{H}^+][\text{OH}^-] \quad (15.2)$$

where K_w is called the **ion-product constant**, which is the *product of the molar concentrations of H⁺ and OH[−] ions at a particular temperature*.

In pure water at 25°C, the concentrations of H⁺ and OH[−] ions are equal and found to be [H⁺] = 1.0 × 10^{−7} M and [OH[−]] = 1.0 × 10^{−7} M. Thus, from Equation (15.2), at 25°C

$$K_w = (1.0 \times 10^{-7})(1.0 \times 10^{-7}) = 1.0 \times 10^{-14}$$

Tap water and water from underground sources do conduct electricity because they contain many dissolved ions.

Recall that in pure water, [H₂O] = 55.5 M (see Section 13.3).

If you could randomly remove and examine 10 particles (H₂O, H⁺, or OH[−]) per second from a liter of water, on average it would take you two years, working nonstop, to find one H⁺ ion!

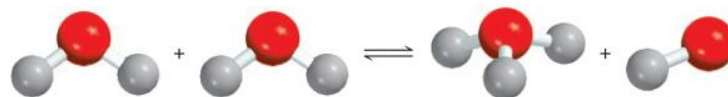


Figure 15.1 Reaction between two water molecules to form hydronium and hydroxide ions.

Whether we have pure water or an aqueous solution of dissolved species, the following relation *always* holds at 25°C:

$$K_w = [\text{H}^+][\text{OH}^-] = 1.0 \times 10^{-14} \quad (15.3)$$

Whenever $[\text{H}^+] = [\text{OH}^-]$, the aqueous solution is said to be neutral. In an acidic solution there is an excess of H^+ ions and $[\text{H}^+] > [\text{OH}^-]$. In a basic solution there is an excess of hydroxide ions, so $[\text{H}^+] < [\text{OH}^-]$. In practice we can change the concentration of either H^+ or OH^- ions in solution, but we cannot vary both of them independently. If we adjust the solution so that $[\text{H}^+] = 1.0 \times 10^{-6} \text{ M}$, the OH^- concentration *must* change to

$$[\text{OH}^-] = \frac{K_w}{[\text{H}^+]} = \frac{1.0 \times 10^{-14}}{1.0 \times 10^{-6}} = 1.0 \times 10^{-8} \text{ M}$$

An application of Equation (15.3) is given in Example 15.2.

Example 15.2

The concentration of OH^- ions in a certain household ammonia cleaning solution is 0.0025 M . Calculate the concentration of H^+ ions.

Strategy We are given the concentration of the OH^- ions and asked to calculate $[\text{H}^+]$. The relationship between $[\text{H}^+]$ and $[\text{OH}^-]$ in water or an aqueous solution is given by the ion-product of water, K_w [Equation (15.3)].

Solution Rearranging Equation (15.3), we write

$$[\text{H}^+] = \frac{K_w}{[\text{OH}^-]} = \frac{1.0 \times 10^{-14}}{0.0025} = 4.0 \times 10^{-12} \text{ M}$$

Check Because $[\text{H}^+] < [\text{OH}^-]$, the solution is basic, as we would expect from the earlier discussion of the reaction of ammonia with water.

Practice Exercise Calculate the concentration of OH^- ions in a HCl solution whose hydrogen ion concentration is 1.3 M .

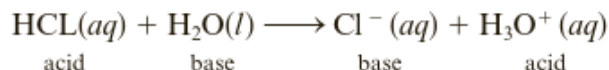
Similar problems: 15.15, 15.16.

► Acid and Base Strengths

The Brønsted–Lowry concept considers an acid–base reaction as a proton-transfer reaction. It is useful to consider such acid–base reactions as a competition between species for protons. From this point of view, you can order acids by their relative strengths as proton donors. The stronger acids are those that lose their protons more easily than other acids. Similarly, the stronger bases are those that hold on to protons more strongly than other bases.

15.4 Relative Strengths of Acids and Bases

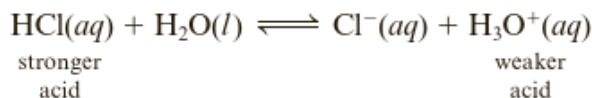
By comparing various acid–base reactions, you can construct a table of relative strengths of acids and bases (see Table 15.2). To see how you might do this, look again at what we have called a strong acid. Recall that an acid is strong if it completely ionizes in water. In the reaction of hydrogen chloride with water, for example, water acts as a base, accepting the proton from HCl.



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The reverse reaction occurs only to an extremely small extent. Because the reaction goes almost completely to the right, you say that HCl is a strong acid. However, even though the reaction goes almost completely to products, you can consider the reverse reaction. In it, the Cl^- ion acts as the base, accepting a proton from the acid H_3O^+ .

Look at this reaction in terms of the relative strengths of the two acids, HCl and H_3O^+ . Because HCl is a strong acid, it must lose its proton readily, more readily than H_3O^+ does. You would say that HCl is a stronger acid than H_3O^+ , or that, of the two, H_3O^+ is the weaker acid.



It is important to understand that the terms *stronger* and *weaker* are used here only in a comparative sense. The H_3O^+ ion is a relatively strong acid.

Table 15.2 Relative Strengths of Acids and Bases

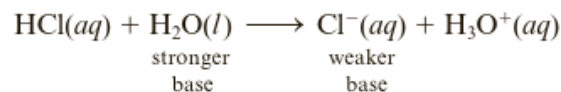
	Acid	Base	
Strongest acids ↓ Weakest acids	HClO ₄	ClO ₄ ⁻	Weakest bases ↑ Strongest bases
	H ₂ SO ₄	HSO ₄ ⁻	
	HI	I ⁻	
	HBr	Br ⁻	
	HCl	Cl ⁻	
	HNO ₃	NO ₃ ⁻	
	H ₃ O ⁺	H ₂ O	
	HSO ₄ ⁻	SO ₄ ²⁻	
	H ₂ SO ₃	HSO ₃ ⁻	
	H ₃ PO ₄	H ₂ PO ₄ ⁻	
	HNO ₂	NO ₂ ⁻	
	HF	F ⁻	
	HC ₂ H ₃ O ₂	C ₂ H ₃ O ₂ ⁻	
	Al(H ₂ O) ₆ ³⁺	Al(H ₂ O) ₅ OH ²⁺	
	H ₂ CO ₃	HCO ₃ ⁻	
	H ₂ S	HS ⁻	
	HClO	ClO ⁻	
	HBrO	BrO ⁻	
	NH ₄ ⁺	NH ₃	
	HCN	CN ⁻	
	HCO ₃ ⁻	CO ₃ ²⁻	
	H ₂ O ₂	HO ₂ ⁻	
	HS ⁻	S ²⁻	
	H ₂ O	OH ⁻	

An acid–base reaction normally goes in the direction of the weaker acid. You can use this fact to compare the relative strengths of any two acids, as we did in comparing the relative strengths of HCl and H₃O⁺. As another example, look at the ionization of acetic acid, HC₂H₃O₂, in water.



Experiment reveals that in a 0.1 *M* acetic acid solution, only about 1% of the acetic acid molecules have ionized by this reaction. This implies that HC₂H₃O₂ is a weaker acid than H₃O⁺. If you look at the similar situation for 0.1 *M* HF, you find that about 3% of the HF molecules have dissociated. Thus, HF is a weaker acid than H₃O⁺ but a stronger acid than HC₂H₃O₂. You have already seen that HCl is stronger than H₃O⁺. Thus, you have determined that the acid strengths for these four acids are in the order HCl > H₃O⁺ > HF > HC₂H₃O₂.

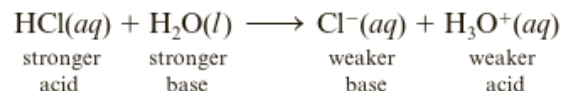
You can also view this same reaction in terms of the bases, H_2O and Cl^- . A stronger base picks up a proton more readily than does a weaker one. Water has greater base strength than the Cl^- ion; that is, H_2O picks up protons more readily than Cl^- does. In fact, Cl^- has little attraction for protons. (If it had more, HCl would not lose its proton so readily.) Because H_2O molecules compete more successfully for protons than Cl^- ions do, the reaction goes almost completely to the right. (Note that the arrow on the right in Table 15.2 points from H_2O to Cl^- .) That is, at the completion of the reaction, concentrations of product species (Cl^- and H_3O^+) are much greater than the concentration of reactant HCl .



The reaction goes in the direction of the weaker base.

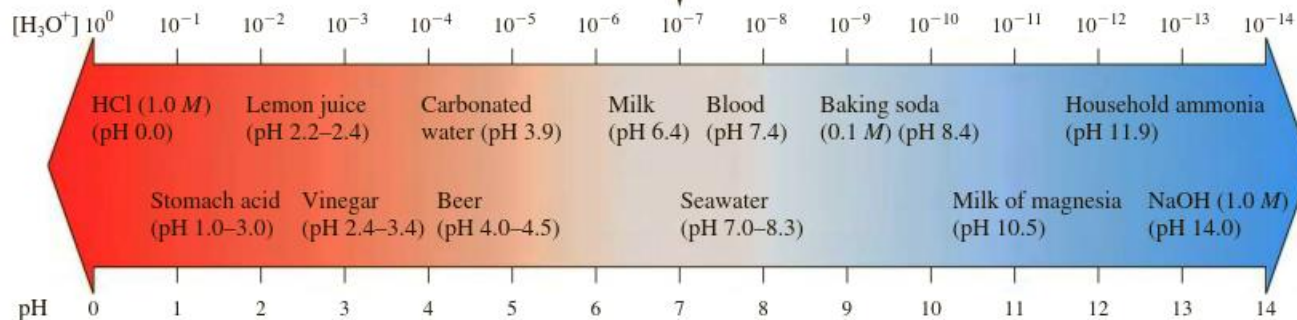
By comparing reactions between different pairs of bases, you can arrive at a relative order for base strengths, just as for acids. A definite relationship exists between acid and base strengths. When you say an acid loses its proton readily, you can also say that its conjugate base does not hold the proton very tightly. *The strongest acids have the weakest conjugate bases, and the strongest bases have the weakest conjugate acids.* This means that a list of conjugate bases of the acids in Table 15.2 will be in order of increasing base strength. The weakest bases will be at the top of the table and the strongest at the bottom.

You can use Table 15.2 to predict the direction of an acid–base reaction. The direction for an acid–base reaction always favors the weaker acid and weaker base; that is, the normal direction of reaction is from the stronger acid and base to the weaker acid and base. For the reaction we have been discussing, you have



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The reaction follows the direction of the arrows at the left and right of Table 15.2.



15.3 pH—A Measure of Acidity

Because the concentrations of H^+ and OH^- ions in aqueous solutions are frequently very small numbers and therefore inconvenient to work with, Søren Sørensen[†] in 1909 proposed a more practical measure called pH. The **pH** of a solution is defined as *the negative logarithm of the hydrogen ion concentration (in mol/L)*:

$$\text{pH} = -\log [\text{H}_3\text{O}^+] \quad \text{or} \quad \text{pH} = -\log [\text{H}^+] \quad (15.4)$$

The pH of concentrated acid solutions can be negative. For example, the pH of a 2.0 M HCl solution is -0.30 .

Keep in mind that Equation (15.4) is simply a definition designed to give us convenient numbers to work with. The negative logarithm gives us a positive number for pH, which otherwise would be negative due to the small value of $[\text{H}^+]$. Furthermore, the term $[\text{H}^+]$ in Equation (15.4) pertains only to the *numerical part* of the expression

[†]Søren Peer Lauritz Sørensen (1868–1939). Danish biochemist. Sørensen originally wrote the symbol as pH and called p the “hydrogen ion exponent” (*Wasserstoffionexponent*); it is the initial letter of Potenz (German), *puissance* (French), and power (English). It is now customary to write the symbol as pH.

for hydrogen ion concentration, for we cannot take the logarithm of units. Thus, like the equilibrium constant, the pH of a solution is a dimensionless quantity.

Because pH is simply a way to express hydrogen ion concentration, acidic and basic solutions at 25°C can be distinguished by their pH values, as follows:

Acidic solutions: $[\text{H}^+] > 1.0 \times 10^{-7} \text{ M}$, $\text{pH} < 7.00$

Basic solutions: $[\text{H}^+] < 1.0 \times 10^{-7} \text{ M}$, $\text{pH} > 7.00$

Neutral solutions: $[\text{H}^+] = 1.0 \times 10^{-7} \text{ M}$, $\text{pH} = 7.00$

Notice that pH increases as $[\text{H}^+]$ decreases.

Sometimes we may be given the pH value of a solution and asked to calculate the H^+ ion concentration. In that case, we need to take the antilog of Equation (15.4) as follows:

$$[\text{H}_3\text{O}^+] = 10^{-\text{pH}} \quad \text{or} \quad [\text{H}^+] = 10^{-\text{pH}} \quad (15.5)$$

Be aware that the definition of pH just shown, and indeed all the calculations involving solution concentrations (expressed either as molarity or molality) discussed in previous chapters, are subject to error because we have implicitly assumed ideal behavior. In reality, ion-pair formation and other types of intermolecular interactions may affect the actual concentrations of species in solution. The situation is analogous to the relationships between ideal gas behavior and the behavior of real gases discussed in Chapter 5. Depending on temperature, volume, and amount and type of gas present, the measured gas pressure may differ from that calculated using the ideal gas equation. Similarly, the actual or “effective” concentration of a solute may not be what we think it is, knowing the amount of substance originally dissolved in solution. Just as we have the van der Waals and other equations to reconcile discrepancies between the ideal gas and nonideal gas behavior, we can account for nonideal behavior in solution.

One way is to replace the concentration term with *activity*, which is the effective concentration. Strictly speaking, then, the pH of solution should be defined as

$$\text{pH} = -\log a_{\text{H}^+} \quad (15.6)$$

where a_{H^+} is the activity of the H^+ ion. As mentioned in Chapter 14 (Section 14.2), for an ideal solution activity is numerically equal to concentration. For real solutions, activity usually differs from concentration, sometimes appreciably. Knowing the solute concentration, there are reliable ways based on thermodynamics for estimating its activity, but the details are beyond the scope of this text. Keep in mind, therefore, that, except for dilute solutions, the measured pH is usually not the same as that calculated from Equation (15.4) because the concentration of the H^+ ion in molarity is not numerically equal to its activity value. Although we will continue to use concentration in our discussion, it is important to know that this approach will give us only an approximation of the chemical processes that actually take place in the solution phase.

In the laboratory, the pH of a solution is measured with a pH meter (Figure 15.2). Table 15.1 lists the pHs of a number of common fluids. As you can see, the pH of body



Figure 15.2 A pH meter is commonly used in the laboratory to determine the pH of a solution. Although many pH meters have scales marked with values from 1 to 14, pH values can, in fact, be less than 1 and greater than 14.

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Table 15.1**The pHs of Some Common Fluids**

Sample	pH Value
Gastric juice in the stomach	1.0–2.0
Lemon juice	2.4
Vinegar	3.0
Grapefruit juice	3.2
Orange juice	3.5
Urine	4.8–7.5
Water exposed to air*	5.5
Saliva	6.4–6.9
Milk	6.5
Pure water	7.0
Blood	7.35–7.45
Tears	7.4
Milk of magnesia	10.6
Household ammonia	11.5

*Water exposed to air for a long period of time absorbs atmospheric CO_2 to form carbonic acid, H_2CO_3 .

fluids varies greatly, depending on location and function. The low pH (high acidity) of gastric juices facilitates digestion whereas a higher pH of blood is necessary for the transport of oxygen. These pH-dependent actions will be illustrated in Chemistry in Action essays in this chapter and Chapter 16.

A pOH scale analogous to the pH scale can be devised using the negative logarithm of the hydroxide ion concentration of a solution. Thus, we define pOH as

$$\text{pOH} = -\log [\text{OH}^-] \quad (15.7)$$

If we are given the pOH value of a solution and asked to calculate the OH^- ion concentration, we can take the antilog of Equation (15.7) as follows

$$[\text{OH}^-] = 10^{-\text{pOH}} \quad (15.8)$$

Now consider again the ion-product constant for water at 25°C :

$$[\text{H}^+][\text{OH}^-] = K_w = 1.0 \times 10^{-14}$$

Taking the negative logarithm of both sides, we obtain

$$\begin{aligned} -(\log [\text{H}^+] + \log [\text{OH}^-]) &= -\log (1.0 \times 10^{-14}) \\ -\log [\text{H}^+] - \log [\text{OH}^-] &= 14.00 \end{aligned}$$

From the definitions of pH and pOH we obtain

$$\text{pH} + \text{pOH} = 14.00 \quad (15.9)$$

Equation (15.9) provides us with another way to express the relationship between the H^+ ion concentration and the OH^- ion concentration.

Example 15.3

The concentration of H^+ ions in a bottle of table wine was $3.2 \times 10^{-4} \text{ M}$ right after the cork was removed. Only half of the wine was consumed. The other half, after it had been standing open to the air for a month, was found to have a hydrogen ion concentration equal to $1.0 \times 10^{-3} \text{ M}$. Calculate the pH of the wine on these two occasions.

Strategy We are given the H^+ ion concentration and asked to calculate the pH of the solution. What is the definition of pH?

Solution According to Equation (15.4), $\text{pH} = -\log [\text{H}^+]$. When the bottle was first opened, $[\text{H}^+] = 3.2 \times 10^{-4} \text{ M}$, which we substitute in Equation (15.4)

$$\begin{aligned}\text{pH} &= -\log [\text{H}^+] \\ &= -\log (3.2 \times 10^{-4}) = 3.49\end{aligned}$$

On the second occasion, $[\text{H}^+] = 1.0 \times 10^{-3} \text{ M}$, so that

$$\text{pH} = -\log (1.0 \times 10^{-3}) = 3.00$$

Comment The increase in hydrogen ion concentration (or decrease in pH) is largely the result of the conversion of some of the alcohol (ethanol) to acetic acid, a reaction that takes place in the presence of molecular oxygen.

Practice Exercise Nitric acid (HNO_3) is used in the production of fertilizer, dyes, drugs, and explosives. Calculate the pH of a HNO_3 solution having a hydrogen ion concentration of 0.76 M .

Similar problems: 15.17, 15.18.

In each case the pH has only two significant figures. The two digits to the right of the decimal in 3.49 tell us that there are two significant figures in the original number (see Appendix 3).

Example 15.5 Calculating the pH from the Hydronium-Ion Concentration

Gaining Mastery Toolbox

Critical Concept 15.5

The pH of a solution equals $-\log [\text{H}^+]$.

Solution Essentials:

- $K_w = [\text{H}_3\text{O}^+][\text{OH}^-] = 1.0 \times 10^{-14}$ at 25°C
- Molar concentration

A sample of orange juice has a hydronium-ion concentration of $2.9 \times 10^{-4} \text{ M}$. What is the pH? Is the solution acidic?

Problem Strategy pH is the negative base 10 logarithm of the molar hydronium-ion concentration. Determine the log using an electronic calculator; instructions vary, so refer to those for your calculator. On many calculators, you enter the H_3O^+ concentration, press *log*, and change the sign of the result. You should write the pH with the same number of decimal places as there are significant figures in $[\text{H}_3\text{O}^+]$.

Solution

$$\text{pH} = -\log [\text{H}_3\text{O}^+] = -\log(2.9 \times 10^{-4}) = 3.54$$

The pH is less than 7.00, so the solution is **acidic** (as you expect for orange juice).

Example 15.4

The pH of rainwater collected in a certain region of the northeastern United States on a particular day was 4.82. Calculate the H^+ ion concentration of the rainwater.

Strategy Here we are given the pH of a solution and asked to calculate $[\text{H}^+]$. Because pH is defined as $\text{pH} = -\log [\text{H}^+]$, we can solve for $[\text{H}^+]$ by taking the antilog of the pH; that is, $[\text{H}^+] = 10^{-\text{pH}}$, as shown in Equation (15.5).

Solution From Equation (15.4)

$$\text{pH} = -\log [\text{H}^+] = 4.82$$

Therefore,

$$\log [\text{H}^+] = -4.82$$

To calculate $[\text{H}^+]$, we need to take the antilog of -4.82 :

$$[\text{H}^+] = 10^{-4.82} = 1.5 \times 10^{-5} M$$

Check Because the pH is between 4 and 5, we can expect $[\text{H}^+]$ to be between $1 \times 10^{-4} M$ and $1 \times 10^{-5} M$. Therefore, the answer is reasonable.

Practice Exercise The pH of a certain orange juice is 3.33. Calculate the H^+ ion concentration.

Similar problem: 15.19.

Example 15.5

In a NaOH solution $[\text{OH}^-]$ is $2.9 \times 10^{-4} M$. Calculate the pH of the solution.

Strategy Solving this problem takes two steps. First, we need to calculate pOH using Equation (15.7). Next, we use Equation (15.9) to calculate the pH of the solution.

Solution We use Equation (15.7):

$$\begin{aligned}\text{pOH} &= -\log [\text{OH}^-] \\ &= -\log (2.9 \times 10^{-4}) \\ &= 3.54\end{aligned}$$

Now we use Equation (15.9):

$$\begin{aligned}\text{pH} + \text{pOH} &= 14.00 \\ \text{pH} &= 14.00 - \text{pOH} \\ &= 14.00 - 3.54 = 10.46\end{aligned}$$

Alternatively, we can use the ion-product constant of water, $K_w = [\text{H}^+][\text{OH}^-]$ to calculate $[\text{H}^+]$, and then we can calculate the pH from the $[\text{H}^+]$. Try it.

Check The answer shows that the solution is basic ($\text{pH} > 7$), which is consistent with a NaOH solution.

Practice Exercise The OH^- ion concentration of a blood sample is $2.5 \times 10^{-7} M$. What is the pH of the blood?

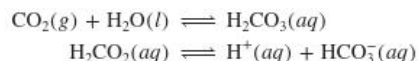
Similar problem: 15.18.

CHEMISTRY *in Action*

Antacids and the pH Balance in Your Stomach

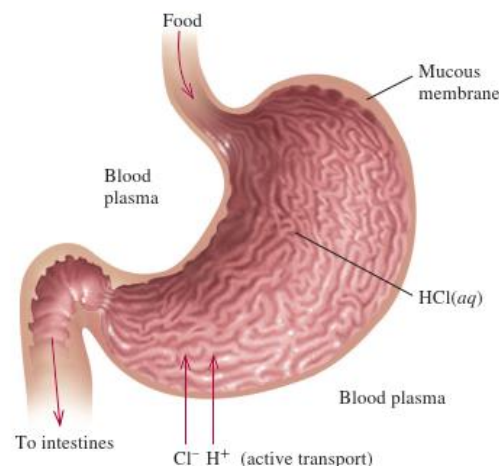
An average adult produces between 2 and 3 L of gastric juice daily. Gastric juice is a thin, acidic digestive fluid secreted by glands in the mucous membrane that lines the stomach. It contains, among other substances, hydrochloric acid. The pH of gastric juice is about 1.5, which corresponds to a hydrochloric acid concentration of 0.03 *M*—a concentration strong enough to dissolve zinc metal! What is the purpose of this highly acidic medium? Where do the H^+ ions come from? What happens when there is an excess of H^+ ions present in the stomach?

A simplified diagram of the stomach is shown here. The inside lining is made up of parietal cells, which are fused together to form tight junctions. The interiors of the cells are protected from the surroundings by cell membranes. These membranes allow water and neutral molecules to pass in and out of the stomach, but they usually block the movement of ions such as H^+ , Na^+ , K^+ , and Cl^- . The H^+ ions come from carbonic acid (H_2CO_3) formed as a result of the hydration of CO_2 , an end product of metabolism:



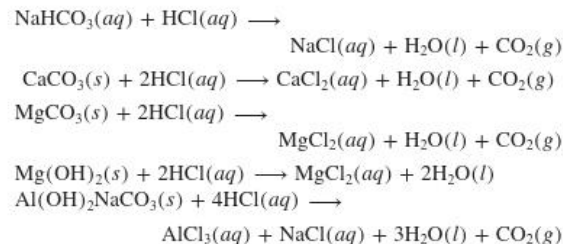
These reactions take place in the blood plasma bathing the cells in the mucosa. By a process known as *active transport*, H^+ ions move across the membrane into the stomach interior. (Active transport processes are aided by enzymes.) To maintain electrical balance, an equal number of Cl^- ions also move from the blood plasma into the stomach. Once in the stomach, most of these ions are prevented from diffusing back into the blood plasma by cell membranes.

The purpose of the highly acidic medium within the stomach is to digest food and to activate certain digestive enzymes. Eating stimulates H^+ ion secretion. A small fraction of these ions normally are reabsorbed by the mucosa, causing many tiny hemorrhages. About half a million cells are shed by the lining every minute, and a healthy stomach is completely relined every three days or so. However, if the acid content is excessively high, the constant influx of H^+ ions through the membrane back to the blood plasma can cause muscle contraction, pain, swelling, inflammation, and bleeding.



A simplified diagram of the human stomach.

One way to temporarily reduce the H^+ ion concentration in the stomach is to take an antacid. The major function of antacids is to neutralize excess HCl in gastric juice. The table “Some Common Commercial Antacid Preparations” lists the active ingredients of some popular antacids. The reactions by which these antacids neutralize stomach acid are as follows:



16.3 Buffer Solutions

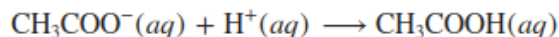
A **buffer solution** is a solution of (1) a weak acid or a weak base and (2) its salt; both components must be present. The solution has the ability to resist changes in pH upon the addition of small amounts of either acid or base. Buffers are very important to chemical and biological systems. The pH in the human body varies greatly from one fluid to another; for example, the pH of blood is about 7.4, whereas gastric juice in the stomach has a pH of about 1.5. These pH values, which are crucial for proper enzyme function and the balance of osmotic pressure, are maintained by buffers in most cases.

A buffer solution must contain a relatively large concentration of acid to react with any OH^- ions that are added to it, and it must contain a similar concentration of base to react with any added H^+ ions. Furthermore, the acid and the base components of the buffer must not consume each other in a neutralization reaction. These requirements are satisfied by an acid-base conjugate pair, for example, a weak acid and its conjugate base (supplied by a salt) or a weak base and its conjugate acid (supplied by a salt).

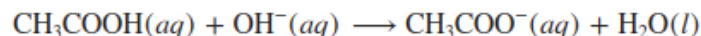
A simple buffer solution can be prepared by adding comparable molar amounts of acetic acid (CH_3COOH) and its salt sodium acetate (CH_3COONa) to water. The equilibrium concentrations of both the acid and the conjugate base (from CH_3COONa) are assumed to be the same as the starting concentrations (see Section 16.2). A solution containing these two substances has the ability to neutralize either added acid or added base. Sodium acetate, a strong electrolyte, dissociates completely in water:



If an acid is added, the H^+ ions will be consumed by the conjugate base in the buffer, CH_3COO^- , according to the equation



If a base is added to the buffer system, the OH^- ions will be neutralized by the acid in the buffer:



As you can see, the two reactions that characterize this buffer system are identical to those for the common ion effect described in Example 16.1. The *buffering capacity*, that is, the effectiveness of the buffer solution, depends on the amount of acid and conjugate base from which the buffer is made. The larger the amount, the greater the buffering capacity.



EACH 100 mL CONTAINS DOBUTAMINE HYDROCHLORIDE USP EQUIVALENT TO 200 mg DOBUTAMINE 5 g DEXTROSE HYDROUS USP 5 mEq/L SODIUM BISULFITE ADDED AS A STABILIZER pH ADJUSTED WITH SODIUM HYDROXIDE AND/OR HYDROCHLORIC ACID pH 3.5 (2.5 TO 5.5) OSMOLARITY 266 mOsmol/L (CALC) STERILE NONPYROGENIC SINGLE DOSE CONTAINER DRUG ADDITIVES SHOULD NOT BE MADE TO THIS SOLUTION DOSAGE INTRAVENOUSLY AS DIRECTED BY A PHYSICIAN SEE DIRECTIONS CAUTIONS MUST NOT BE USED IN

Fluids for intravenous injection must include buffer systems to maintain the proper blood pH.

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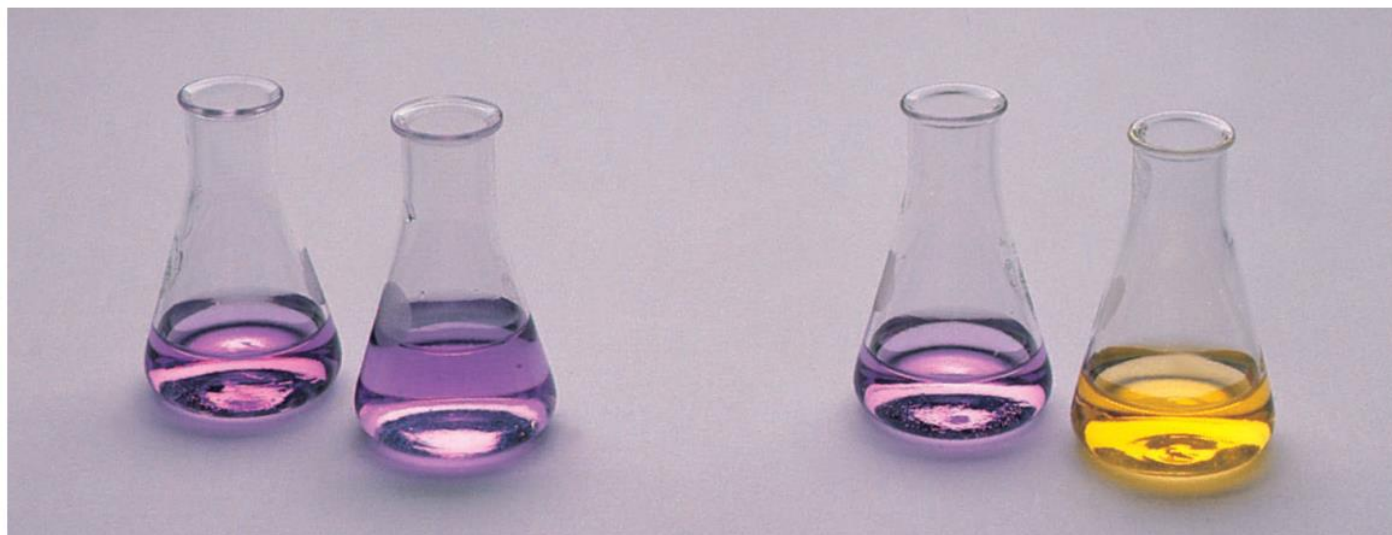


Figure 16.1 The acid-base indicator bromophenol blue (added to all solutions shown) is used to illustrate buffer action. The indicator's color is blue-purple above pH 4.6 and yellow below pH 3.0. (a) A buffer solution made up of 50 mL of 0.1 M CH_3COOH and 50 mL of 0.1 M CH_3COONa . The solution has a pH of 4.7 and turns the indicator blue-purple. (b) After the addition of 40 mL of 0.1 M HCl solution to the solution in (a), the color remains blue-purple. (c) A 100-mL CH_3COOH solution whose pH is 4.7. (d) After the addition of 6 drops (about 0.3 mL) of 0.1 M HCl solution, the color turns yellow. Without buffer action, the pH of the solution decreases rapidly to less than 3.0 upon the addition of 0.1 M HCl.

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Example 16.2

Which of the following solutions can be classified as buffer systems? (a) $\text{KH}_2\text{PO}_4/\text{H}_3\text{PO}_4$, (b) $\text{NaClO}_4/\text{HClO}_4$, (c) $\text{C}_5\text{H}_5\text{N}/\text{C}_5\text{H}_5\text{NHCl}$ ($\text{C}_5\text{H}_5\text{N}$ is pyridine; its K_b is given in Table 15.4). Explain your answer.

Strategy What constitutes a buffer system? Which of the preceding solutions contains a weak acid and its salt (containing the weak conjugate base)? Which of the preceding solutions contains a weak base and its salt (containing the weak conjugate acid)? Why is the conjugate base of a strong acid not able to neutralize an added acid?

Solution The criteria for a buffer system is that we must have a weak acid and its salt (containing the weak conjugate base) or a weak base and its salt (containing the weak conjugate acid).

- (a) H_3PO_4 is a weak acid, and its conjugate base, H_2PO_4^- , is a weak base (see Table 15.5). Therefore, this is a buffer system.
- (b) Because HClO_4 is a strong acid, its conjugate base, ClO_4^- , is an extremely weak base. This means that the ClO_4^- ion will not combine with a H^+ ion in solution to form HClO_4 . Thus, the system cannot act as a buffer system.
- (c) As Table 15.4 shows, $\text{C}_5\text{H}_5\text{N}$ is a weak base and its conjugate acid, $\text{C}_5\text{H}_5\text{NH}^+$ (the cation of the salt $\text{C}_5\text{H}_5\text{NHCl}$), is a weak acid. Therefore, this is a buffer system.

16.4 Acid-Base Titrations

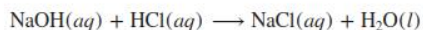
Having discussed buffer solutions, we can now look in more detail at the quantitative aspects of acid-base titrations (see Section 4.6). We will consider three types of reactions: (1) titrations involving a strong acid and a strong base, (2) titrations involving a weak acid and a strong base, and (3) titrations involving a strong acid and a weak base. Titrations involving a weak acid and a weak base are complicated by the hydrolysis of both the cation and the anion of the salt formed. It is difficult to determine the equivalence point in these cases. Therefore, these titrations will not be dealt with here. Figure 16.3 shows the arrangement for monitoring the pH during the course of a titration.



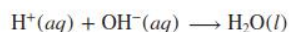
Figure 16.3 A pH meter is used to monitor an acid-base titration.
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Strong Acid–Strong Base Titrations

The reaction between a strong acid (say, HCl) and a strong base (say, NaOH) can be represented by



or in terms of the net ionic equation



Consider the addition of a 0.100 M NaOH solution (from a buret) to an Erlenmeyer flask containing 25.0 mL of 0.100 M HCl. For convenience, we will use only three significant figures for volume and concentration and two significant figures for pH. Figure 16.4 shows the pH profile of the titration (also known as the titration curve). Before the addition of NaOH, the pH of the acid is given by $-\log(0.100)$, or 1.00. When NaOH is added, the pH of the solution increases slowly at first. Near the equivalence point the pH begins to rise steeply, and at the equivalence point (that is, the point at which equimolar amounts of acid and base have reacted) the curve rises almost vertically. In a strong acid–strong base titration both the hydrogen ion and hydroxide ion concentrations are very small at the equivalence point (approximately $1 \times 10^{-7} M$); consequently, the addition of a single drop

of the base can cause a large increase in $[\text{OH}^-]$ and in the pH of the solution. Beyond the equivalence point, the pH again increases slowly with the addition of NaOH.

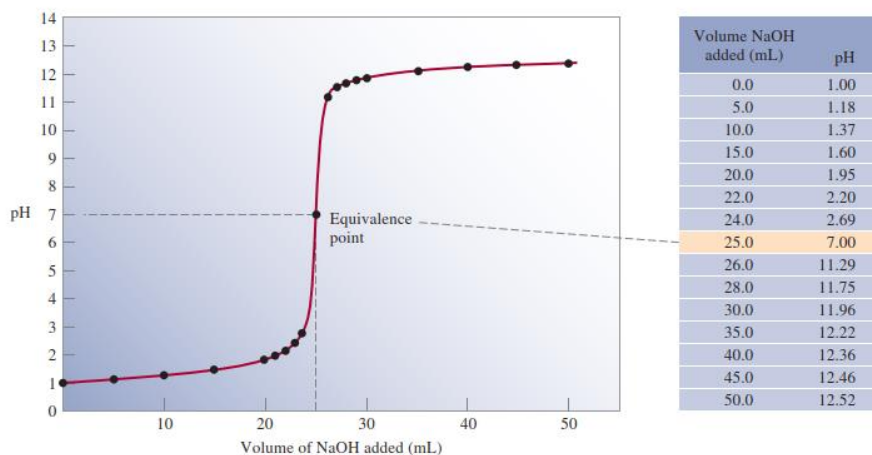
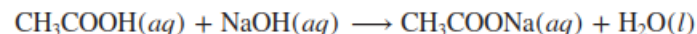


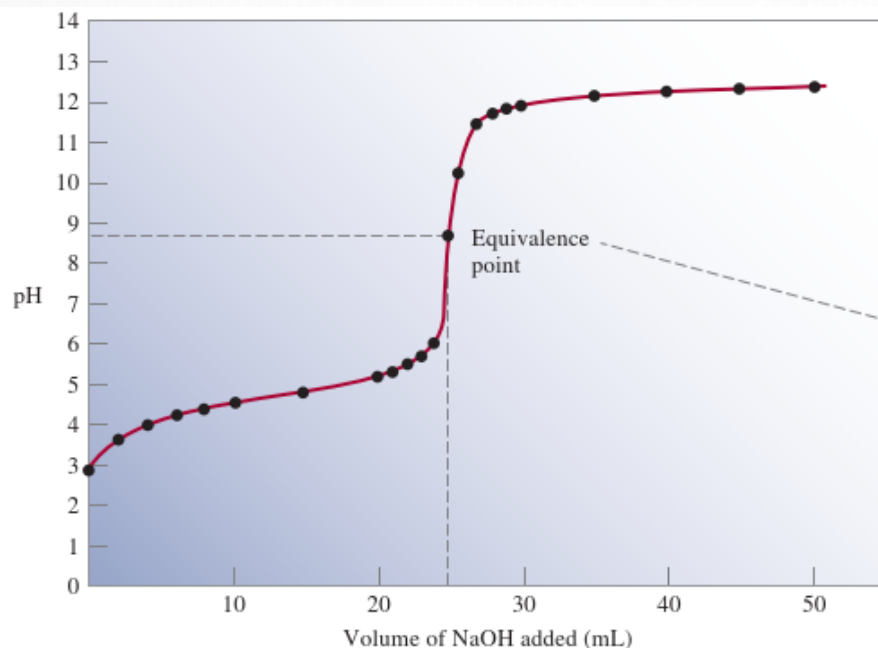
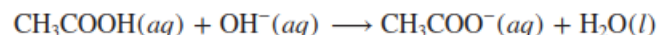
Figure 16.4 pH profile of a strong acid–strong base titration. A 0.100 M NaOH solution is added from a buret to 25.0 mL of a 0.100 M HCl solution in an Erlenmeyer flask (see Figure 4.21). This curve is sometimes referred to as a titration curve.

Weak Acid–Strong Base Titrations

Consider the neutralization reaction between acetic acid (a weak acid) and sodium hydroxide (a strong base):



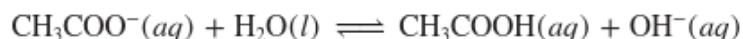
This equation can be simplified to



Volume NaOH added (mL)	pH
0.0	2.87
5.0	4.14
10.0	4.57
15.0	4.92
20.0	5.35
22.0	5.61
24.0	6.12
25.0	8.72
26.0	10.29
28.0	11.75
30.0	11.96
35.0	12.22
40.0	12.36
45.0	12.46
50.0	12.52

Figure 16.5 pH profile of a weak acid–strong base titration. A 0.100 M NaOH solution is added from a buret to 25.0 mL of a 0.100 M CH_3COOH solution in an Erlenmeyer flask. Due to the hydrolysis of the salt formed, the pH at the equivalence point is greater than 7.

The acetate ion undergoes hydrolysis as follows:



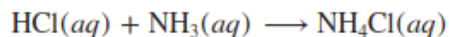
Therefore, at the equivalence point, when we only have sodium acetate present, the pH will be *greater than 7* as a result of the excess OH^- ions formed (Figure 16.5). Note that this situation is analogous to the hydrolysis of sodium acetate (CH_3COONa) (see Section 15.10).

Student Hot Spot

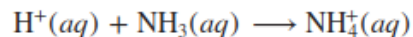
Student data indicate you may struggle with weak acid–strong base titrations. Access the SmartBook to view additional Learning Resources on this topic.

Strong Acid–Weak Base Titrations

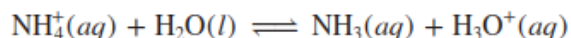
Consider the titration of HCl, a strong acid, with NH_3 , a weak base:



or simply



The pH at the equivalence point is *less than 7* due to the hydrolysis of the NH_4^+ ion:



or simply



Because of the volatility of an aqueous ammonia solution, it is more convenient to add hydrochloric acid from a buret to the ammonia solution. Figure 16.6 shows the titration curve for this experiment.

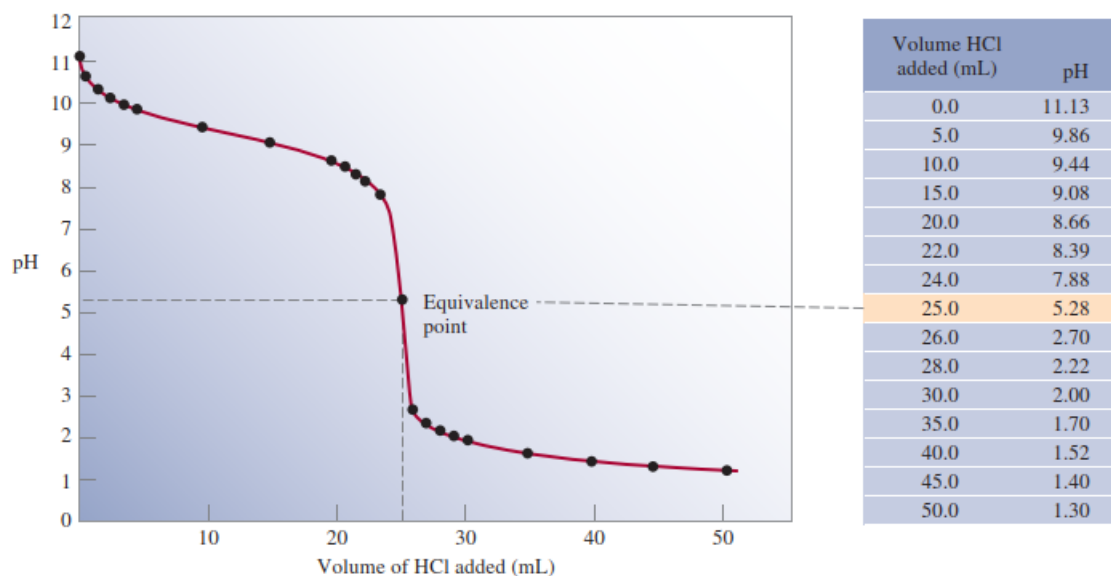


Figure 16.6 pH profile of a strong acid–weak base titration. A 0.100 M HCl solution is added from a buret to 25.0 mL of a 0.100 M NH_3 solution in an Erlenmeyer flask. As a result of salt hydrolysis, the pH at the equivalence point is lower than 7.